

When Nurses Meet Vibe-Coding: Shifting from Problem Talk to Situated Specification in Participatory Design

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Abstract

Participatory Design (PD) empowers end-users as co-designers, yet contemporary practice often reduces PD to ideation workshops where users contribute ideas but remain excluded from implementation. This gap stems from an asymmetry in technical knowledge—users cannot experience how their ideas translate into working systems, limiting feedback to abstract problem framing. We conducted a 110-minute workshop with five clinical nurses designing an EMR interface, incorporating real-time “vibe coding”—LLM-assisted prototyping that transforms verbal descriptions into functional interfaces within minutes. Through thematic analysis of utterances and post-session surveys, we observed shifts in design articulation: from abstract problem statements toward situated specifications referencing concrete screen elements, conditional triggers, and failure modes. Participants reported enhanced agency, identifying as co-creators rather than passive informants. We contribute a methodological template for integrating vibe coding into PD and preliminary observations suggesting that real-time implementation may help surface tacit domain knowledge and reconnect implementation with genuine participation.

CCS Concepts

• **Human-centered computing** → **Participatory design**; *User studies*; • **Computing methodologies** → Natural language processing.

Keywords

Participatory Design, Vibe Coding, Real-Time Prototyping, User Engagement, Co-Design

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1 Introduction

Participatory Design (PD) empowers end-users as active co-designers rather than passive informants, fostering mutual learning between designers and domain experts [13, 19, 27]. In healthcare contexts, PD has been particularly valuable for designing Electronic Medical Record (EMR) systems that align with practitioners’ situated needs [5, 10]. However, contemporary PD practice often falls short of its democratic ideals. A systematic review of 185 PD cases from SIGCHI conferences reveals that most studies narrowly equate PD with ideation workshops, where users contribute sticky notes and sketches but remain excluded from actual implementation [12, 21]. The fundamental problem is an asymmetry in technical knowledge: users can articulate problems verbally, but cannot directly experience how their ideas translate into working systems—a dynamic that risks deskilling domain expertise by reducing it to abstract input [22]. Consequently, their feedback remains abstract, anchored to general frustrations rather than specific interface behaviors, lacking what Schon [26] calls “reflection-in-action.”

Recent advances in generative AI, particularly Large Language Models (LLMs) capable of producing working code from natural language descriptions, present an opportunity to restructure this dynamic. The emerging practice of “vibe coding,” which uses LLMs to rapidly generate executable artifacts during design sessions, compresses the ideation-to-implementation cycle to minutes, enabling users to see their verbal descriptions materialize as interactive interfaces in real time [18]. While researchers have begun exploring AI-mediated design processes, including AI as a collaborative partner in creative work, generative AI combined with speculative co-design techniques, and image-generation as a boundary object [9, 28, 31], these studies focus on AI-generated artifacts rather than

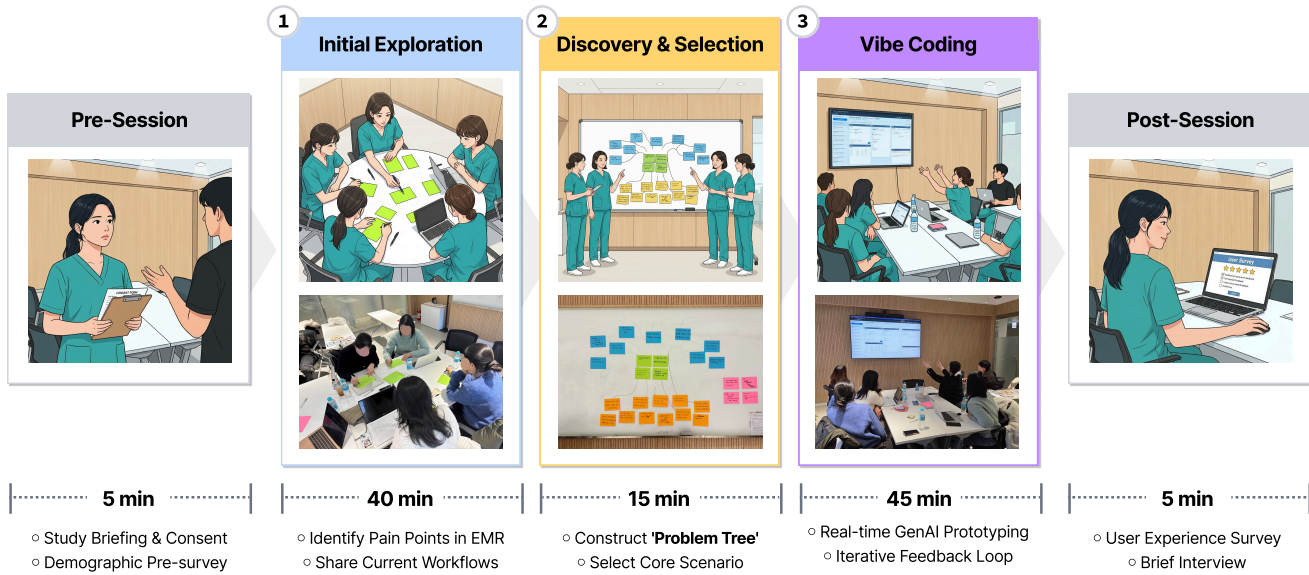


Figure 1: Overview of the 110-minute participatory design workshop with five nurses. Phases 1–2 employed traditional PD methods (group discussion, Problem Tree) to surface abstract design goals, while Phase 3 introduced vibe coding to translate these goals into executable artifacts in real time.

functional systems. Whether LLM-based live coding can enable domain experts to participate in building working software remains underexplored.

To investigate this potential, we conducted a 110-minute participatory design workshop with five nurses designing an EMR interface, incorporating real-time vibe coding. The session progressed from Initial Exploration of workflow pain points, through Discovery and Selection of priority design goals, to Vibe Coding where a developer implemented participants' ideas in real time—allowing us to trace idea re-specification through live screen-level implementation. Thematic analysis of transcripts suggested that abstract concerns articulated in early phases—information integration, summarization, and automation scope—tended to be re-specified through interaction with executable artifacts, as participants surfaced tacit procedural knowledge (i.e., practical know-how about how work actually unfolds in specific clinical situations) about safety constraints, ward-specific norms, and appropriate automation boundaries that had not emerged during earlier abstract discussions. Post-session surveys indicated positive responses across all dimensions, with particularly high ratings on agency and role shift (i.e., participants describing a shift from giving feedback to actively shaping system behavior), as participants reported feeling like "co-creators" rather than passive users.

This work offers preliminary observations on how participants' design articulations shifted from abstract problem statements toward situated specifications (i.e., concrete screen-level decisions tied to specific constraints and use scenarios) when interacting with executable artifacts. We also provide a methodological template and open questions for integrating vibe coding into participatory design practice.

2 Methods

Participants. We recruited five registered nurses (all female, 5+ years clinical experience) from tertiary hospitals through professional networks. All participants were domain experts with deep familiarity with existing EMR systems and daily workflow challenges. Participants received 50 USD compensation for their participation.

Procedure. The IRB-approved workshop comprised 110 minutes structured to first surface domain problems through traditional PD methods, then translate selected problems into executable artifacts through vibe coding (Figure 1). The **Pre-Session** (5 minutes) involved study briefing, informed consent, and a demographic pre-survey capturing professional background and AI familiarity. **Phase 1**, Initial Exploration (40 minutes), focused on identifying pain points in current EMR workflows. Participants engaged in facilitated group discussion, sharing specific examples of documentation challenges and handover inefficiencies, with emerging themes captured on sticky notes arranged on a shared whiteboard. **Phase 2**, Discovery and Selection (15 minutes), structured the identified issues using Problem Tree methodology. Participants collaboratively mapped root causes and downstream consequences, then voted to select a core scenario—nursing handover documentation—for subsequent prototyping. **Phase 3**, Vibe Coding (45 minutes), introduced real-time GenAI prototyping to implement the prioritized design goals. A developer used GPT-5.1-Codex-Max with Cursor IDE to translate participants' verbal ideas into functional interfaces, with both the development environment and the resulting prototype projected on a shared screen. When participants expressed requests such as "add a feature that generates handover notes from patient information," working interfaces appeared within minutes, and participants refined their specifications based on what they

Phases 1–2: Abstract Articulation	Phase 3: Participant Re-specification	Triggering Artifact (Vibe Coding)	Resulting Situated Specification
Integration. Fragmented information across EMR systems hinders rapid patient understanding; need to “integrate” dispersed records.	Suggested a flexible, chat-based interface to support information integration.	Floating clinical reasoning assistant chat.	Integration reframed as an interface-allocation and evidence-navigation problem , emphasizing source linking, scoped highlighting, and verifiability rather than aggregation.
Summarization & Standardization. Extensive, heterogeneous records increase cognitive load; need for summarization and common handover structures across wards.	Synthesized summarization and standardization into an AI-assisted handover organized using an SBAR-P template.	AI-assisted handover interface structured around SBAR-P.	The instantiated SBAR-P structure exposed limits of one-size-fits-all standardization , motivating ward-specific adaptation and omission.
Automation Scope. Desire for AI assistance to reduce documentation burden and improve efficiency.	Articulated constraints on over-automation, preferring bounded assistance aligned with temporal scope and genre norms.	Incrementally enhanced prototype with structured generation, source linking, and highlighting.	Automation shifted from full draft generation to scoped text improvement , focusing on refining nurses’ own terse notes within explicit constraints.

Table 1: Tracing how design ideas articulated abstractly in Phases 1–2 appeared in more situated forms during Phase 3, following the introduction of executable artifacts.

observed. While participants shaped feature requests, interaction logic, and automation boundaries, architectural decisions (e.g., data structures, API integration, error handling) remained with the developer to ensure feasibility and stability within the workshop timeframe. To maintain transparency, the developer (1) paraphrased each request back to participants for confirmation before coding, (2) presented alternative implementations when multiple options existed, and (3) kept all prompts visible on the shared screen. The **Post-Session** (5 minutes) included a user experience survey and brief semi-structured interviews capturing reflections on the workshop experience. All sessions were video-recorded and transcribed for analysis.

Measures. Survey data assessed participants’ vibe coding experience across five dimensions using 7-point Likert scales (two items each; Figure 2): Efficiency and Speed, Iterative Design, Quality and Alignment, Psychological Safety, and Agency and Role Shift. Transcript data captured all design ideas articulated by participants, categorized by phase to compare ideation patterns before and after prototype introduction.

Analysis. Survey responses were analyzed descriptively using diverging stacked bar charts. Transcripts underwent thematic analysis [2–4]. Two researchers independently coded all Phase 1–3 utterances, focusing on shifts from abstract problem framing to screen-grounded design reasoning. Codes were iteratively refined through discussion to develop higher-level themes. Utterances were further categorized by their functional role in the design process (e.g., abstract problem articulation, screen-grounded re-specification of prior ideas, artifact-driven reaction). Design ideas were compared by specificity—whether they specified interaction triggers, user roles, and articulated limits of automation.

3 Findings

This section presents two main findings from the workshop: how executable prototypes transformed abstract problem discussions into situated design specifications, and how screen-grounded interaction shifted participants’ roles from users to active co-designers.

3.1 Tracing Idea Transformation Across Phases: From Problem Enumeration to Situated Re-specification

Table 1 documents how concerns articulated during Phases 1–2 reappeared in Phase 3, often with different characteristics. The three traces presented below reflect higher-level themes developed through thematic coding of Phase 1–3 utterances. Rather than generating new ideas, Phase 3 systematically converted prior abstract desiderata (e.g., integration, summarization, standardization) into situated specifications—constraints, scoping rules, and domain-normative distinctions—through interaction with executable artifacts.

Across all three traces, executable prototypes functioned not as solution endpoints but as probes that bridged abstract expectations and practical specification. For integration, a general desire to “bring information together” was re-specified as a need for navigable allocation and verifiable evidence traversal. For summarization and standardization, initially separate concerns converged into a request for AI-assisted handover generation, which in turn prompted reassessment of the limits of one-size-fits-all templates and the need for ward-specific adaptation. Finally, as automation capabilities accumulated, participants articulated boundaries on what AI should attempt, shifting preferred assistance from full draft generation toward scoped text improvement grounded in existing notes.

Taken together, these traces show that Phase 3 did not simply elicit additional feedback but appeared to shift the nature of participation itself. When encountering executable artifacts, participants articulated tacit procedural knowledge about safety, genre norms, and attention management that had not surfaced in earlier abstract discussions.

3.2 Participant Role Shift: From Users to Co-Designers

Post-workshop survey results showed positive agreement across all five dimensions (Figure 2), but participants’ verbal accounts revealed experiential transitions not fully captured by quantitative measures.

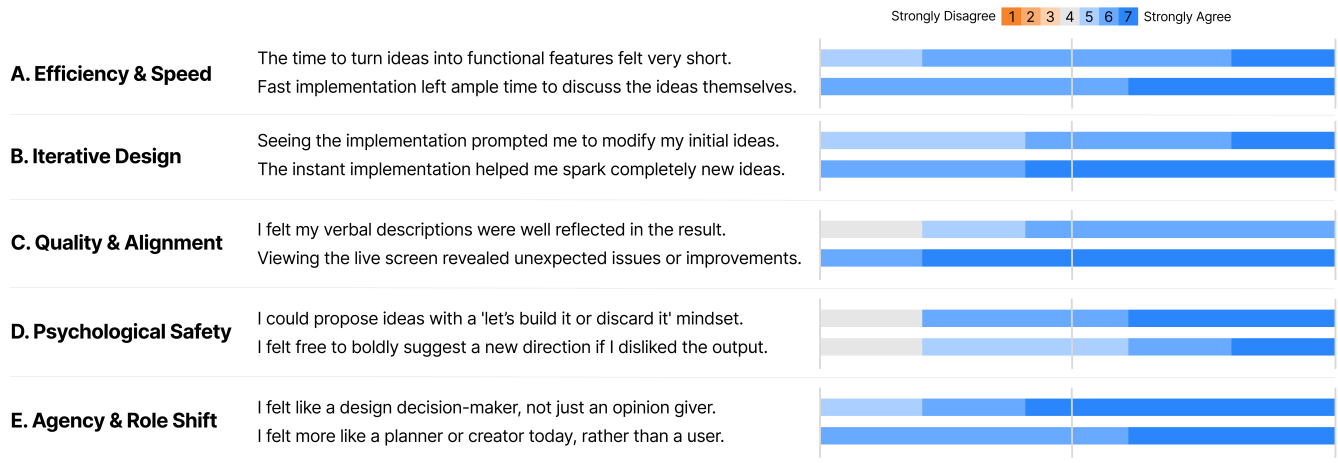


Figure 2: Post-workshop survey responses on vibe coding experience (N=5). All five dimensions showed positive consensus, with Agency & Role Shift receiving the highest ratings (M=6.4)—participants reported feeling like co-creators rather than passive informants.

One of the most salient findings was a cognitive shift reported by participants when moving from abstract discussion to screen-grounded interaction. Reflecting on their experience across the phases, P4 explained how sense-making initially felt difficult without visual artifacts, but became more concrete once live interaction was introduced: “At first, without anything visual, I didn’t know where to begin. But once we worked with this, I started suggesting changes and giving feedback right away.” This account helps explain why discussions situated solely in an abstract problem space tend to remain limited, and contextualizes the survey result indicating that participants *revised their initial ideas after seeing the implementation* (M = 5.8, SD = 0.84).

Theoretically, the most meaningful results emerged in the dimension of **Agency and Role Shift**. All participants rated above 5 on both statements: *I felt like a design decision-maker, not just an opinion giver* (M = 6.4, SD = 0.89) and *I felt more like a planner or creator than a user* (M = 6.4, SD = 0.55). When asked whether they imagined themselves actually using the system while discussing it, P5 immediately agreed: “Yes, exactly”. These responses suggest that vibe coding did not merely elicit ideas, but prompted participants to actively simulate future system use. P3 summarized this experience succinctly as “*designing together felt intuitive*”.

Participants also offered constructive critiques regarding methodological limitations. P2 noted that the prototype’s distance from everyday practice made it difficult to anchor discussion in concrete improvement strategies: “*If this had been a bit more similar to the systems I use now, more specific ideas might have come up.*” This feedback suggests that the effectiveness of vibe coding may depend on the *fidelity* of the initial prototype, pointing to the need for systematic investigation into baseline design strategies that better align with participants’ professional contexts.

4 Discussion

This section reflects on the implications of our findings for participatory design. We discuss how Vibe Coding can reconnect implementation with participation and outline a methodological design space for future configurations of Vibe Coding-enabled participatory design.

4.1 Reconnecting Implementation and Participation

A long-standing limitation of traditional participatory design lies in asymmetries of technical knowledge: while users can propose ideas, they are typically unable to participate in implementation [21]. As a result, user contributions often remain at an abstract level, and key decisions that arise during implementation are handled outside the participatory space. Although participatory design has always emphasized iteration and co-decision-making in principle, practical constraints in research settings—such as limited time, resources, and technical capacity—often make it difficult to repeatedly implement and refine working systems with participants. In practice, this has led many PD studies to rely on discussion, sketches, or delayed implementation, leaving the implementation phase largely detached from participation.

By lowering the barrier to implementation, vibe coding may help reconnect the implementation phase with participation, enabling concrete system behavior to surface users’ situated reasoning and implicit judgments during the design process. Prior work has begun to explore the use of generative AI for live or rapid prototyping in user-centered design contexts, primarily framing Vibe Coding and related techniques as tools for increasing efficiency in ideation and prototyping, or as methodological infrastructures for running live-prototyping studies [15, 25].

Beyond these benefits, our findings suggest an additional role for Vibe Coding in participatory design. Many of the insights in

our study that we found most notable did not emerge when participants articulated feature requests, but when they encountered and reacted to implemented functionality. Expressions such as realizing that a feature would not work in practice, refining conditional triggers, or anticipating failure scenarios surfaced only in response to artifacts. This pattern suggests that lowering the barrier to implementation may do more than speed up the design process. It may transform implementation into a participatory lens through which users' implicit procedural knowledge and situated reasoning become visible.

We note that the LLM-generated prototype was used as an exploratory probe to elicit situated critique, not as a validated or production-ready clinical system. Questions of debugging transparency, maintainability, and code verification remain open, particularly in safety-critical domains. Moreover, our study design does not allow us to isolate the effects of Vibe Coding itself from other factors such as workshop familiarization, group rapport building, or the presence of any prototype regardless of creation method. As a next step, we plan to conduct controlled comparative studies that contrast Vibe Coding with alternative prototyping configurations, including static mockups and pre-built prototypes [16]. These studies will allow us to more precisely examine when and how Vibe Coding supports the articulation of situated knowledge and re-specification in participatory design.

4.2 Methodological Design Space for Vibe Coding–Enabled Participatory Design & Future Work

As an ongoing line of research, this work begins to surface a broader methodological design space for Vibe Coding–enabled participatory design across different configurations.

One important dimension concerns **the characteristics of the application domain and task**. Vibe Coding may be particularly valuable in contexts shaped by complex workflows and implicit procedural practices [8, 20]. When workflows are complex, verbal descriptions tend to collapse conditional branches that only become visible when participants encounter concrete system behaviors. Similarly, implicit procedural rules that users take for granted are rarely articulated until a working artifact prompts situated critique [29, 30]. These effects may further vary depending on users' familiarity with AI and technical systems, shaping whether live coding primarily supports learning about technological affordances or deeper forms of co-design and critique.

A further methodological dimension concerns **how Vibe Coding roles and responsibilities are distributed among stakeholders**, including users, designers, developers, and researchers. Prior work has examined how AI-mediated development can reshape or obscure domain expertise within technical systems [22]. In our study, researchers primarily performed the live coding, while users shaped the system through situated critique and decision-making. Alternative configurations—such as users or designers performing Vibe Coding with researchers acting as facilitators—may redistribute design authority [11, 23], but also introduce challenges related to cognitive load, coordination, and uneven technical expertise. Rather than treating “who codes” as a binary distinction,

we suggest understanding Vibe Coding as a “third space” [19] facilitating mutual learning [14] and addressing the “symmetry of ignorance” [7].

Finally, another methodological choice concerns **whether Vibe Coding begins from a blank slate or a baseline system**. While starting from scratch may encourage open-ended exploration through generative techniques [24], starting from familiar systems can better support concrete and actionable critique. This observation aligns with the *seeding* approach [6], in which an initial artifact serves as a catalyst for user feedback. Baselines themselves can take different forms, ranging from deliberately crude implementations intended to provoke criticism [1, 17] to systems closely resembling current practice, pointing to opportunities for future comparative studies on how baseline design strategies shape participatory dynamics. Taken together, these dimensions suggest that Vibe Coding is best understood as a methodological approach whose effects depend on how it is configured.

As an exploratory study conducted in a single domain (nursing) with a small sample ($N = 5$) drawn from a single national context (South Korea), the generalizability of our findings is limited. Within these constraints, our observations point to the potential usefulness of Vibe Coding–enabled participatory design in domains characterized by complex workflows and tacit procedural knowledge, where executable prototypes may help participants move beyond abstract discussion and articulate requirements through interaction with working artifacts. In future work, we plan to examine Vibe Coding across a broader range of domains, participant populations, and methodological configurations.

5 Conclusion

Through a 110-minute workshop with five nurses designing an EMR interface, we explored how vibe coding—LLM-assisted real-time prototyping—may influence participatory design dynamics. We observed that executable artifacts appeared to function as probes associated with the conversion of abstract desiderata into situated specifications, surfacing tacit procedural knowledge about safety and automation boundaries that remained latent in earlier discussions. Survey results suggested enhanced agency, with participants identifying as co-creators rather than passive informants. We contribute a methodological template for integrating vibe coding into PD and preliminary evidence that real-time implementation may help reconnect implementation with genuine participation.

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A Demographic Information

Table 2: Demographic Information of Study Participants.

<i>ID</i>	<i>Gender</i>	<i>Working Experience</i>	<i>Clinical Department(s)</i>
P1	F	10+ years	Nephrology and Endocrinology Ward, Nursing Education Team, Surgical Ward, Next-Generation EMR Team
P2	F	5+ years	Intensive Care Unit (ICU)
P3	F	5+ years	Medical Oncology, Hematology, Obstetrics and Gynecology, Integrated Internal Medicine, Emergency Department, etc.
P4	F	5+ years	Integrated Internal Medicine, Hematology, Gynecology, Cardiology, Pulmonology, Mental Health Surgery Open Ward
P5	F	5+ years	Hematology, Gynecology, Cardiology, Surgical Ward